中熔电气
SINOFUSE

PPU-2XL Series

产品规格书

PRODUCT SPECIFICATION

SINOFUSE

成为卓越的能量保护系统解决方案服务商
为整个人类的能量转换活动的安全运行提供保障
Provide excellent integrated solution of energy protection systems
Safeguard energy conversion processes for mankind

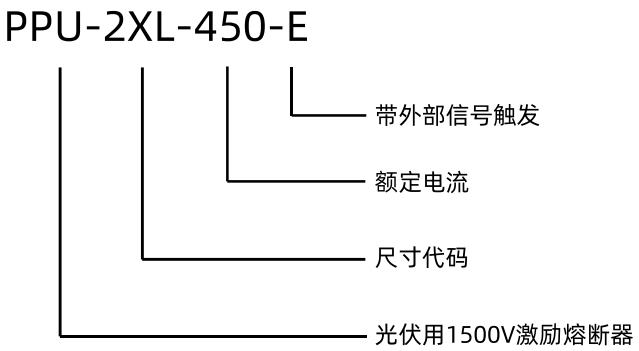
文件变更履历表/ Revision Record

文件名称/ File name：PPU-2XL

文件编号/ File No.：SNC/YC-0030

Version	Content	Execution Date	Author
S0	新制定/First Issue	2025-11-04	Diao Wenjia
S1	补充分断能力描述/Additional explanations on breaking capacity:	2026-01-12	Diao Wenjia

产品命名规则/ Product Naming Rules



产品技术参数/ Product Technical Parameters

- 参考标准/Conform to:IEC 60269
- 额定电压/Rated voltage:DC 1500V
- 额定电流/Rated current:400A～450A
- 分断能力/Breaking capacity:0~30kA[a]
- 符合/Compliant:RoHS
- 认证/Certification:TUV

No.	Part Number	In (A)	I^2t/ (A^2s)		Loss (W)	Resistance Value mΩ (20℃)	Weight (g)	Max Package (pcs)	Remark
			Prearc	Clearing					
1	PPU-2XL-400-E	400	116000	160000	52	0.175~0.215	1290	8	M10 21±1N·m
2	PPU-2XL-450-E	450	135000	182000	60	0.157~0.193			

注：以上I^2t为基于IEC60269-6测试条件下的典型数据，测试电压DC1520V，预期电流30.2kA，时间常数2.75ms。

Note: The above I^2t values represent typical data obtained under IEC 60269-6 test conditions, with a test voltage of DC 1520 V, an expected current of 30.2 kA, and a time constant of 2.75 ms.

产品特性/ Product Features

本产品为光伏全范围保护用激励熔断器。具有低I^2t，低功耗、宽分断范围的特点；主动触发性能参考IEC60269-6要求设计，可通过外部信号触发；用于保护和隔离光伏汇流箱,以及保护其他适合的电路系统。

This product is an incentive fuse for comprehensive photovoltaic protection. It features low I^2t, low power consumption, and a wide breaking range. Its active triggering performance is designed to meet IEC 60269-6 requirements and can be triggered by external signals. It is intended for protecting and isolating photovoltaic combiner boxes, as well as safeguarding other suitable circuit systems.

注：本规格书中所有试验数据均为标准环境下测得，试验参数和接线参考IEC 60269（GB/T 13539）。

[a]：自触发分断能力

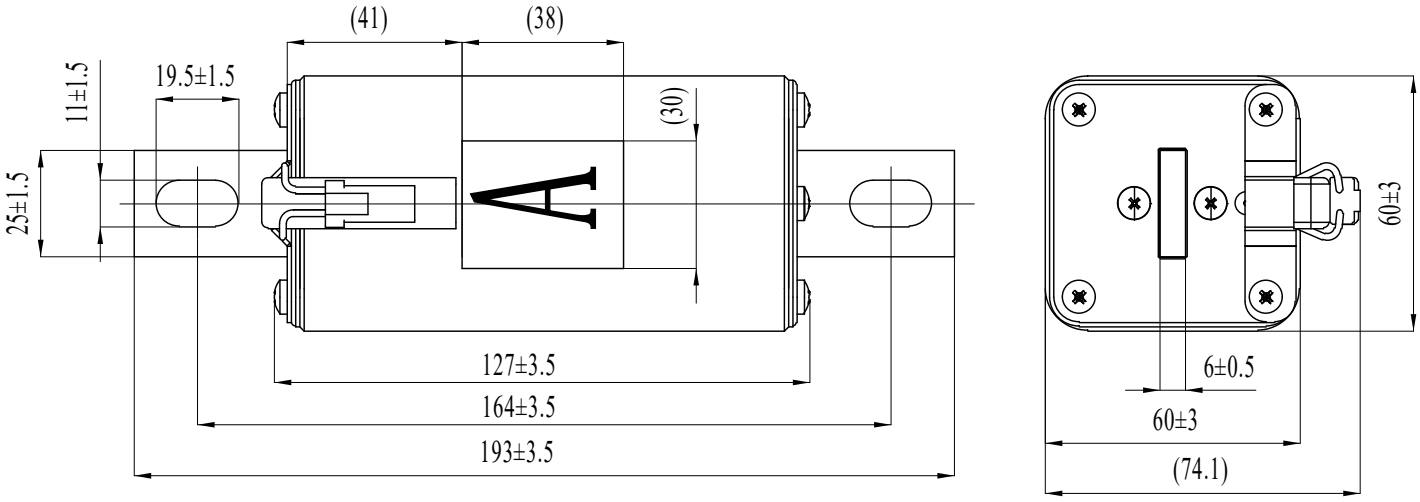
测试电压：≦1520V；预期电流1.5 ~30kA，时间常数2ms；预期电流1.0In ~1.5kA,电感量≤2mH.

Note: All test data in this specification were obtained under standard environmental conditions. Test parameters and wiring arrangements refer to IEC 60269 (GB/T 13539).

[a]: Self-trigger breaking capacity

Test voltage: ≤1520V; Expected current: 1.5 to 30kA, time constant 2ms;Expected current: 1.0In to 1.5kA, inductance ≤2mH.

产品外形尺寸/Dimensions(mm)

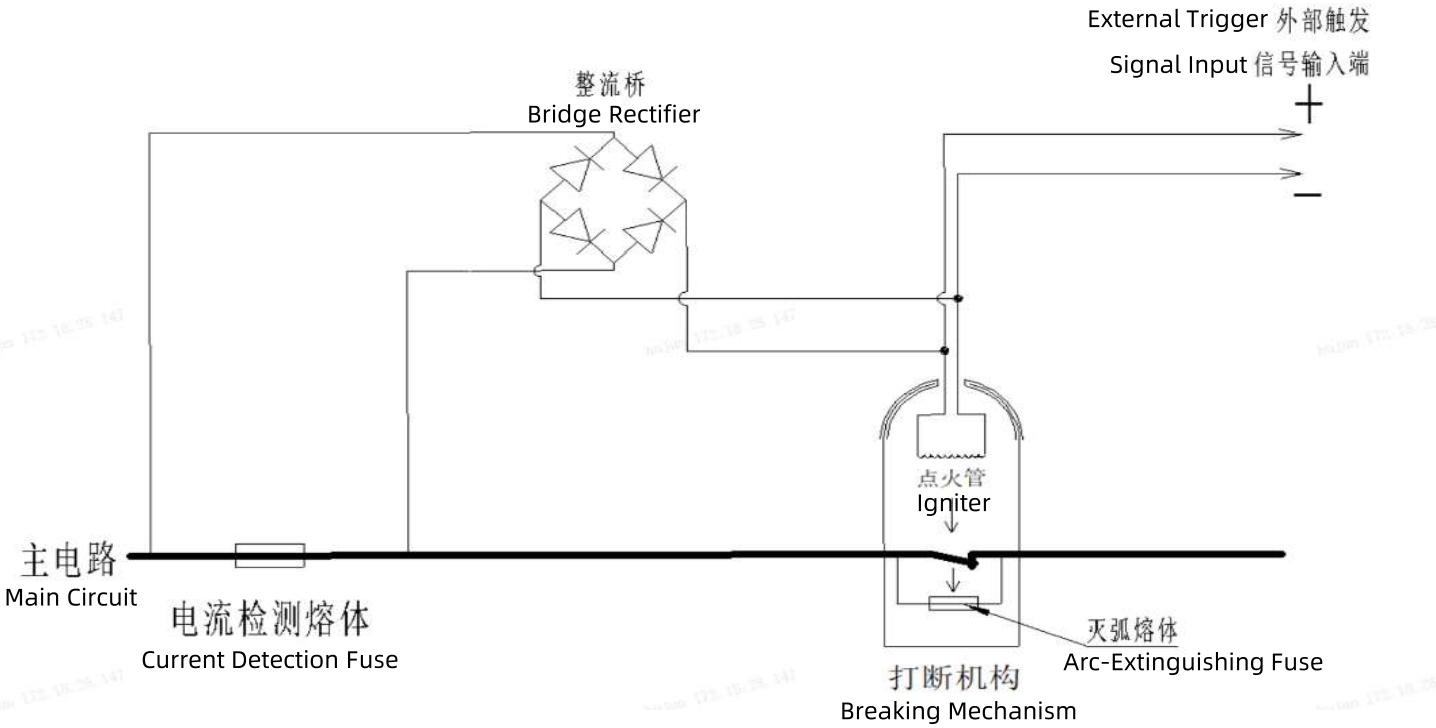


注：两触刀安装面共面度≤1。/Note：Coplanarity of the Two Contact Blade Mounting Surfaces ≤1.

主要材料清单/Main Materials List

No	品名/Product	材料/Material	Remark
1	瓷管/Ceramic tube	瓷/Ceramic	
2	触刀/Terminal	铜/Copper	镀锡/Tin Plating

原理简介/ Introduction to Principles



当主电路出现故障电流一定时间后，电流检测熔体熔断，从而在熔体两端产生一定的电压信号，电压信号通过与之连接的导线经由整流桥触发点火管起爆，点火管起爆后推动切断装置打断主电路导电排和灭弧熔体，从而实现在短时间内切断电路的功能。也可通过外部直接提供信号触发点火管起爆从而实现切断电路。

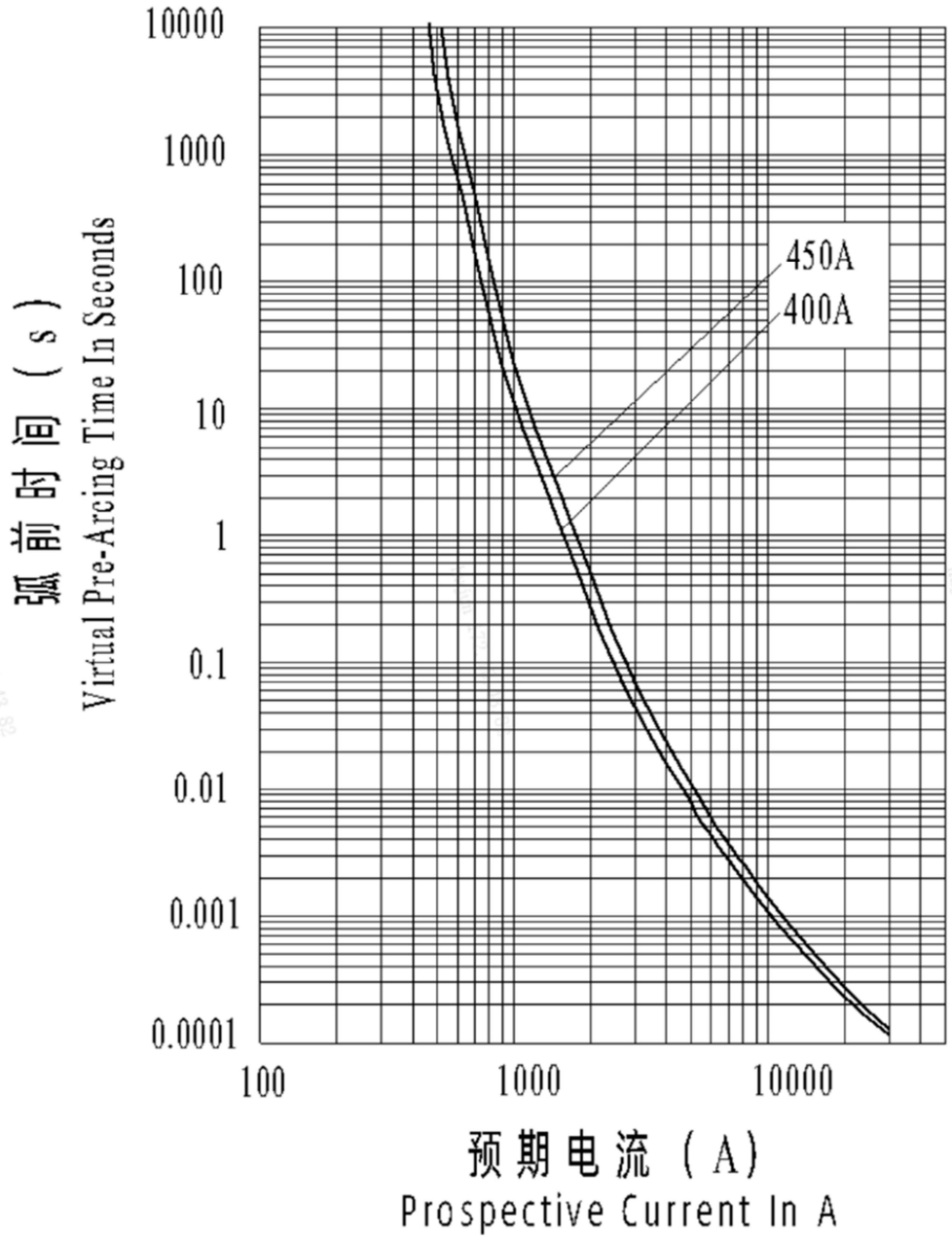
Upon the occurrence of a fault current in the main circuit for a certain duration, the current-sensing fuse melts, thereby generating a voltage signal across its terminals. This voltage signal travels through the connected wiring and, via the rectifier bridge, triggers the ignition tube to detonate. Following detonation, the ignition tube actuates the disconnection mechanism, severing the main circuit's conductive busbar and arc-extinguishing fuse. This achieves the function of interrupting the circuit within a short timeframe. Alternatively, the circuit may be disconnected by directly supplying an external signal to trigger the ignition tube's detonation.

点火器技术条件/ Technical Specifications for Igniters

Project	Standard	Indicators
点火器阻抗/Igniter impedance	符合AK-LV16标准/ Complies with the AK-LV16 standard	1.7-2.3Ω(-40℃-85℃)
点火器检测电流/ Igniter detection current		≦100 mA
触发参数/Triggering parameters		1.75 A / 0.5 ms or 1.2 A/2.0 ms
不触发参数/ No Triggering parameters		0.4A/≤10S
环境温度/Ambient temperature		-40℃-120℃

产品特性曲线/ Characteristic Curve

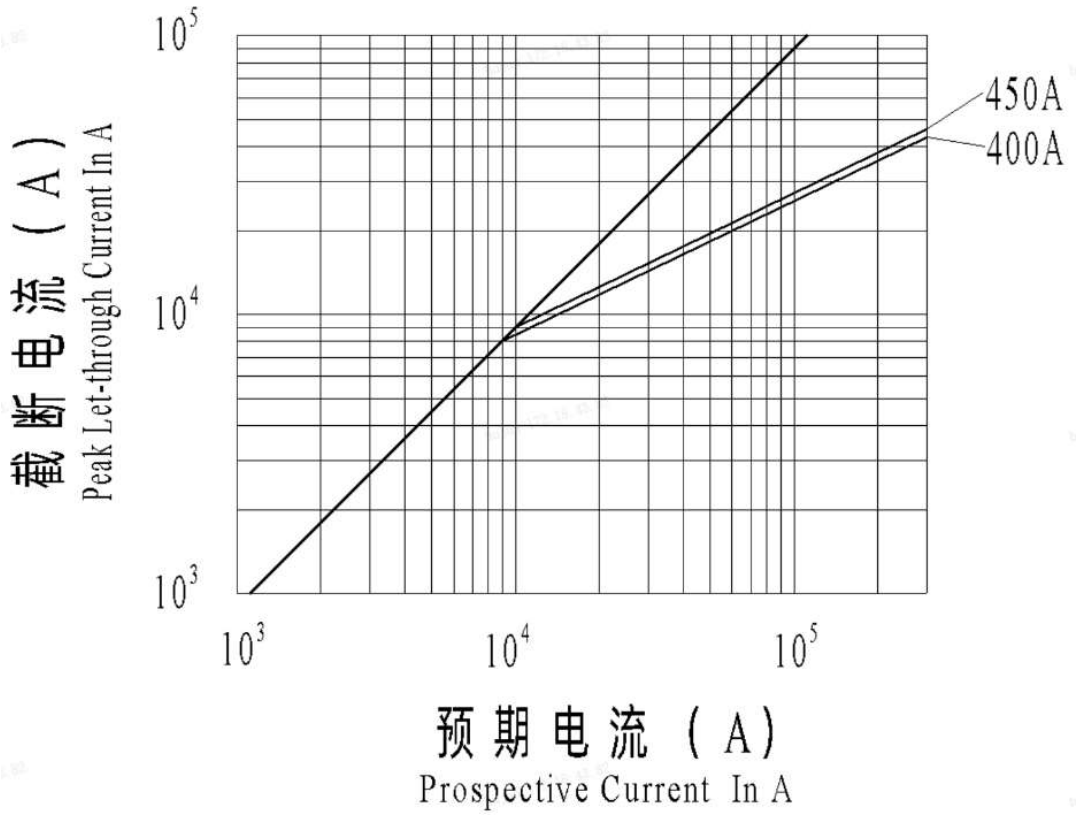
自触发弧前时间-电流特性曲线/ Pre-trigger arc time-current characteristic curve



注/Note:

1. 特性曲线精度为预期电流方向误差±15%;/ Curves: current tolerance ±15%.
2. 特性曲线100ms以下为等效时间，时间常数2ms;/ Below 100ms in the curve is the equivalent pre-arc time，Time constant 2ms.
3.典型燃弧时间≤5ms(参考值，不作为合格判定标准)。/
Typical arcing time ≤ 5 ms (reference value, not to be used as pass/fail criterion).

自触发截断特性曲线/ Self-triggering Truncation Characteristic Curve



注/Note：
截断曲线数据：时间常数为2ms。/Truncation curve data: time constant is 2ms.

运输、存储/Transportation, Storage

运输/Transportation

避免过度挤压或冲击：在运输过程中，需要避免对熔断器施加过度挤压或直接冲击，这可能会损坏熔断器的外壳、接线端子或内部元件，从而影响其性能和安全性。运输过程中应避免雨雪侵袭和机械损伤。

Avoid excessive compression or shock: During transportation, it is necessary to avoid applying excessive compression or direct shock to the fuse. This may damage the fuse housing, terminals or internal components, thus affecting their performance and safety. Avoid rain, snow and mechanical damage during transportation.

防潮防湿：熔断器应远离潮湿和湿润的环境，以防止受潮和腐蚀。必要时，可以使用防潮包装材料或采取其他措施来保护熔断器免受潮湿影响。

Moisture and humidity protection: Fuses should be kept away from damp and wet environments to prevent moisture and corrosion. If necessary, moisture-proof packing materials may be used or other measures taken to protect the fuse from moisture.

存储/Storage

1.产品存储温度：-40℃～85℃；/Storage Temperature: -40℃～85℃.
2.包装存储温度：-40℃～70℃；/Package Temperature: -40℃～70℃.
3.产品存储湿度：相对湿度＜90%，无凝露；/Product Storage Humidity: RH＜90%, no dewing.
4.包装存储湿度：相对湿度＜90%，无凝露。/Package Humidity: RH＜90%, no dewing.

使用条件/Use Conditions

熔断器所有试验均在标准测试条件下进行，因实际使用条件和标准测试条件存在差异，在选用时需考虑其影响。当实际使用条件满足允许使用条件时，需按照对应修正曲线对熔断器的额定值进行修正；当实际使用条件不满足允许使用条件时，请咨询我司。

All tests for the fuse are conducted under standard testing conditions, there is the difference between the actual use conditions and the test conditions, the influence of which should be taken into account when selecting the fuses. When actual usage condition meets tolerable range, the rated value should be corrected according to the corresponding correction curve. Otherwise, please consult Sinofuse.

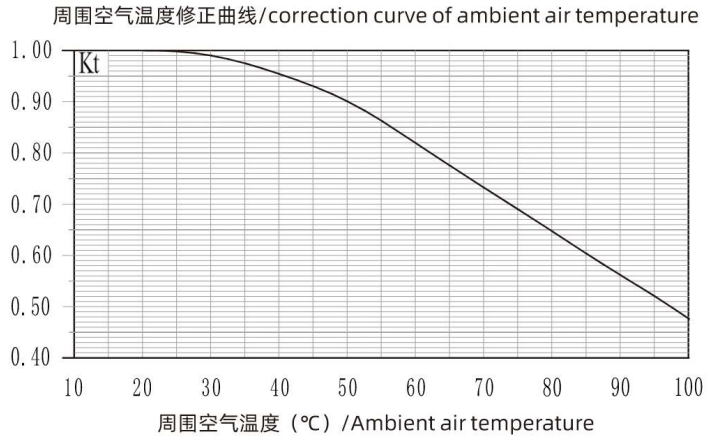
大气条件/Atmosphere

相对湿度<85%，无凝露。/Relative humidity <85%, no condensation

周围空气温度/Ambient Air Temperature

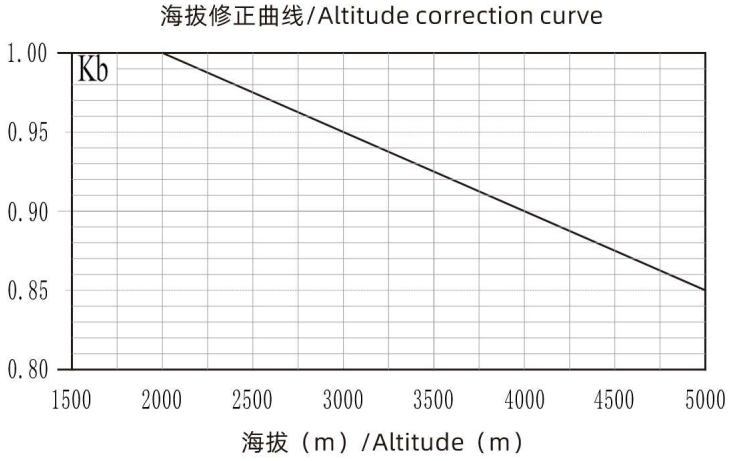
标准典型测试条件：20℃；
Standard Typical Testing Condition：20℃.
允许使用条件：-40℃～75℃，熔断器在周围空气温度20℃以上工作，额定电流按照下面曲线进行修正；温度修正时一般不考虑增加熔断器额定电流，除非20℃以上不是工作范围。

Tolerable Testing Conditions：-40℃～75℃; the rated current of fuse should be corrected according to the following curves when ambient temperature is above 20℃; temperature derating typically does not consider increasing the fuse's rated current，unless above 20℃ is not within the operating range.



海拔/Altitude

标准测试条件：海拔≤2000m；
Standard Testing: Altitude≤2000m.
允许使用条件：海拔≤5000m，不同海拔下熔断器额定电流按照下面曲线进行修正。
Tolerable Range: Altitude≤5000m; rated current should be corrected according to the following curve at different altitudes.



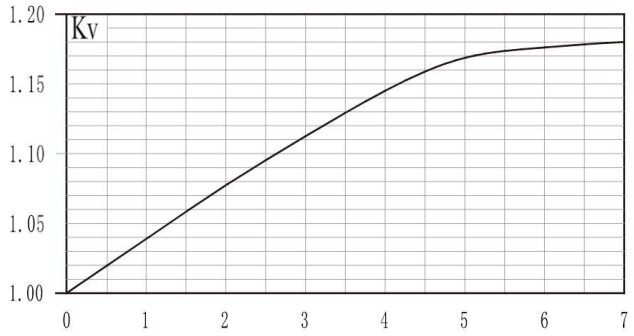
风冷/Air Cooling

标准测试条件：无风，周围1m以内无发热器件；
Standard Testing: no ventilation, no heat source within 1m.
允许使用条件：允许有风及周围有发热器件，但需考虑其影响；
Tolerable Range: ventilation and heat source allowed, but their impacts should be considered.

强制风冷：采用强制风冷可以改善熔断器的散热，提升熔断器的额定电流，强制风冷条件下熔断器额定电流按照下面曲线进行修正。

Forced Air Cooling：Forced air cooling improves the heat dissipation of fuses, enabling an increase in their rated current. Under forced air cooling conditions. Rated current should be corrected according to the following curve.

强制冷却风速修正曲线/Correction curve of forced cooling air velocity



强制冷却风速(m/s)/Forced cooling air velocity(m/s)

铜导线/铜排尺寸/Copper wire/Busbar size

标准测试条件：铜导线/铜排尺寸满足标准要求；

Standard Testing: The size of the copper wire/busbar complies with the standard requirements.

允许使用条件：推荐铜导体截面不小于标准截面的50%（标准截面以实际通流值为前提），其他材质导体需换算为铜导体的尺寸进行校核。

Tolerable Range: It is recommended that the cross-sectional area of the copper conductor should not be less than 50% of the standard cross-sectional area (the standard cross-section is based on actual current-carrying capacity). For conductors made of other materials, their dimensions must be converted to equivalent copper conductor dimensions for verification.

标准导线选用表（参考GB/T13539.1）/Standard Conductor Selection Chart（GB/T13539.1）

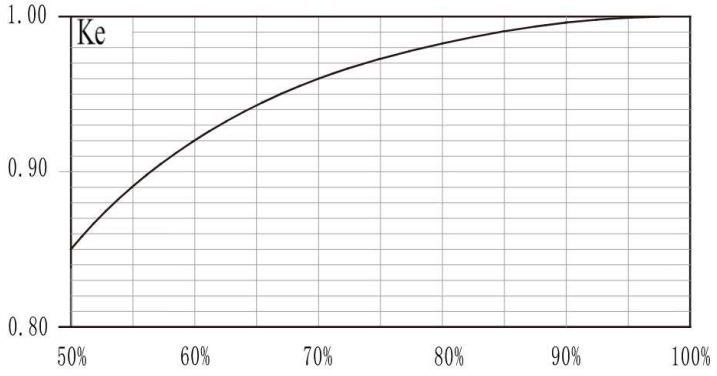
额定电流 A Rated Current	导线截面积 mm² Conductor Cross-sectional Area
2、4、6	1
8、10、12	1.5
16、20	2.5
25	4
32	6
40、50	10
63	16
80	25
100	35
125	50
160	70

额定电流 A Rated Current	导线截面积 mm² Conductor Cross-sectional Area
200、225	95
250、300	120
315、350	185
400、425	240
500	300
630	370
800	480
1000	600
1250	800
> 1250	（1.0~1.6）A/mm²

不同铜导体尺寸下熔断器的额定电流按照下面曲线进行修正

Rated current should be corrected according to the following curve

接线修正曲线/ Wiring correction curve



实际截面/标准截面/ Actual cross-section/standard cross-section

耐受机械振动、冲击能力/Resistance to Mechanical Shock and Vibration

机械冲击满足GB/T28046.3第4.2.1条，振动满足下表参数：

Mechanical shock shall comply with Clause 4.2.1 of GB/T 28046.3, whilst vibration shall meet the parameters specified in the table below:

Vibration Mode	Vibration Direction	Sweep Frequency	Number of Cycles	Vibration Duration	Amplitude	Acceleration
Sweep Vibration	X	10Hz-150Hz-10Hz	10	150min	0.15mm	20m/s²
	Y	10Hz-150Hz-10Hz	10	150min	0.15mm	20m/s²
	Z	10Hz-150Hz-10Hz	10	150min	0.15mm	20m/s²

污染等级/Pollution Level

熔断器满足污染等级2（详见GB 14048.1-2023 7.1.3.2）。

Fuse meets pollution level II (see GB 14048.1-2023 7.1.3.2 for details).

安装共面度/Mounting Coplanarity

软连接安装落差≤8mm，若刚性平面安装，需与厂家沟通。

For flexible connection installation, the vertical displacement shall be≤8mm. If installing on a rigid flat surface, consultation with the manufacturer is required.